QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION

III SEMESTER

Major - MG3DSEPHY207 - Properties of Matter

(2024 ADMISSION ONWARDS)

Duration: 1.5 hrs **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1.	What are the three types of elastic moduli?	[K][CO1]
2.	Why does a perfectly plastic body not obey Hooke's law?	[U][CO1]
3.	What is Poisson’s ratio?	[K][CO2]
4.	How do stress and strain differ?	[U][CO2]
5.	Estimate the Poisson's ratio of a rubber band 15 cm long and 3 cm wide, elongated by 20% and narrowed by 9.5%.	[A][CO2]
6.	Give the SI unit of surface tension.	[K][CO3]
7.	Why do small insects float on water?	[U][CO3]
8.	A box with a mass of 10 kg and a cross-sectional area of 0.5 m^2 is falling through air with a density of 2 kg/m^3. Find its terminal velocity. (Assume the drag coefficient to be 0.5 and take g=10 ms^{-2}.)	[A][CO3]
9.	Consider two identical pipes carrying water and petrol, evaluate which flow is more likely to be turbulent.	[An][CO3]
10.	What do you mean by sonic boom?	[U][CO4]
11.	Define a longitudinal wave. Give two examples.	[U][CO5]
12.	Why sound is a longitudinal wave?	[A][CO5]
13.	What is ultrasound imaging?	[K][CO6]
14.	Two adjacent piano keys are struck simultaneously. The notes emitted by them have frequencies 412 Hz and 417 Hz. Write down the number of beats heard per second.	[U][CO6]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15.	Describe stress-strain curve for a ductile material.	[U][CO1]
16.	How do torsion pendulum helps in determining moment of inertia of a circular disc?	[U][CO2]
17.	One end of a rope of length 5 m and diameter 8 mm hanging from an iron bar, where the other end is fixed. A weight of 100 N is attached to the free end. Then what will be the elongation of the rope. Young's modulus of the material is $2x10^{11}$ N/m^2	[A][CO2]
18.	Describe the excess pressure inside spherical and cylindrical drops?	[U][CO3]
19.	Assuming that the surface tension of rain water is 0.072 N/m. Find the difference of pressure between inside and outside of a rain drop of diameter 0.02 cm.	[A][CO3]
20.	A wave is described by $y(x,t)=0.02 \sin(3x-4t)$. Find the wave number and angular frequency.	[U][CO4]
21.	Describe the differential equation of a one-dimensional wave. What are its applications?	[U][CO5]
22.	How is ultrasonic used in flaw detection?	[U][CO6]

[5x6 = 30]